

Assignment 2: Exploratory Data Analysis

$$1. m(a+bx) = \frac{1}{N} \sum_{i=1}^N (a+bx_i) = \frac{1}{N} \sum_{i=1}^N \underset{\text{constant}}{a} + \frac{1}{N} \sum_{i=1}^N \underset{\text{constant}}{bx_i} = \frac{1}{N} \cdot (a \cdot N) + b \cdot \underbrace{\left(\frac{1}{N} \sum_{i=1}^N x_i \right)}_{m(x)} = \boxed{a + b \cdot m(x)}$$

$$2. \text{COV}(X, a+bY) = \frac{1}{N} \sum_{i=1}^N (x_i - m(x)) (a+by_i - \underbrace{m(a+by)}_{\substack{\text{from 1, this is } a+b \cdot m(y)}}) = \frac{1}{N} \sum_{i=1}^N (x_i - m(x)) (a+by_i - a - b \cdot m(y)) \\ = \frac{1}{N} \sum_{i=1}^N (x_i - m(x)) (by_i - b \cdot m(y)) \\ = b \cdot \underbrace{\frac{1}{N} \sum_{i=1}^N (x_i - m(x)) (y_i - m(y))}_{\text{COV}(X, Y)} \\ = \boxed{b \cdot \text{COV}(X, Y)}$$

$$3. \text{COV}(a+bx, a+bx) = \frac{1}{N} \sum_{i=1}^N \underbrace{(a+bx_i - m(a+bx))}_{a+b \cdot m(x)} \underbrace{(a+bx_i - m(a+bx))}_{a+b \cdot m(x)} \\ = \frac{1}{N} \sum_{i=1}^N (a+bx_i - a - b \cdot m(x)) (a+bx_i - a - b \cdot m(x)) \\ = \frac{1}{N} \sum_{i=1}^N (bx_i - b \cdot m(x)) (bx_i - b \cdot m(x)) \\ = b \cdot b \cdot \underbrace{\frac{1}{N} \sum_{i=1}^N (x_i - m(x)) (x_i - m(x))}_{\text{COV}(X, X)} \\ = \boxed{b^2 \text{COV}(X, X)}$$

$$\text{COV}(X, X) = \frac{1}{N} \sum_{i=1}^N (x_i - m(x)) (x_i - m(x)) = \frac{1}{N} \sum_{i=1}^N \underbrace{(x_i - m(x))^2}_{s^2} = \boxed{s^2}$$

4. No, it is not always true. It only holds true if the function $g(\cdot)$ is linear.

For example, if $g(\cdot) = x^2$, and let X be 0, 2.

$$m(g(X)) = g(m(X))$$

$$m(X^2) = g\left(\frac{0+2}{2}\right) \leftarrow \text{counterexample}$$

$$\frac{0+2^2}{2} = 1^2$$

$$2 \neq 1$$